

Modal Force and its realization across languages

Omar Agha
University of Maryland

Paloma Jeretič
University of Pennsylvania

Abstract

Force is one of the two major dimensions of meaning along which modals in natural language vary, the other being flavor. In this chapter, we provide an overview of formal approaches to modal force, with a focus on cross-linguistic data and typological evidence.

In formal semantics, a common assumption is that necessity and possibility modals express (respectively) universal and existential quantification over possible worlds. However, there are some modals that do not fall into this binary in an obvious way. In this chapter, we explore two such cases: weak necessity modals, which are universal-like, but weaker than strong necessity modals, and variable force modals, which can take on universal-like or existential-like meanings depending on features of their linguistic context. We also briefly discuss other constructions which arguably involve covert variable force modal operators, namely imperatives, generics, and bare conditionals. We focus on the analysis of modal force in formal semantics and pragmatics. For a broader perspective on types of modal meanings across languages, see 1.5. EMPIRICAL FOUNDATIONS OF MODALITY.

Some other categories of modal meaning lie outside the possibility-necessity binary, but are discussed in other sections of this handbook. Gradable modality can also be categorized as a third type of modal force that falls outside of the usual possibility-necessity dichotomy, but that is the topic of 3.2. MODALITY, GRADABILITY, AND PROBABILITY, so we do not discuss it here. Ability modals exhibit a more complicated force profile modal, and a detailed discussion can be found in 2.3. CIRCUMSTANTIAL AND ABILITY MODALS.

Each section in this chapter introduces a particular empirical modal force phenomenon and its associated analyses. In Section 1, we introduce the distinction between possibility and (strong) necessity. In Section 2, we discuss weak necessity modals. In Section 3, we give an overview of variable force modal items. In Section 4, we discuss force in other modal constructions.

1 Possibility and necessity

Modal meanings are typically divided between necessity and possibility.

- | | | | |
|-----|----|---------------------------------|-----------|
| (1) | a. | Jockl must be the murderer. | $\forall$ |
| | b. | Jockl might be the murderer. | $\exists$ |
| (2) | a. | You are required to wear a hat. | $\forall$ |
| | b. | You are allowed to wear a hat. | $\exists$ |

As noticed by Aristotle in *On Interpretation*, necessity and possibility modals are duals. They are interdefinable via negation, allowing for the following equivalences between the (a) and (b) sentences.

- | | | | |
|-----|----|---|---|
| (3) | a. | Jockl must be the murderer. | $\forall$ |
| | b. | It's not the case that Jockl might not be the murderer. | $\neg\exists\neg \Leftrightarrow \forall$ |
| (4) | a. | You are not required to wear a hat. | $\neg\forall$ |
| | b. | You are allowed to not wear a hat. | $\exists\neg \Leftrightarrow \neg\forall$ |

In formal semantics, the standard way to treat the duality facts is to analyze necessity modals as universal quantifiers over worlds, and possibility modals as existential quantifiers. On this analysis, the equivalences in (3) and (4) follow from DeMorgan's laws for quantifiers.

Just like quantifiers in the nominal domain, modals quantify over a restricted domain of possible worlds. Following foundational work by Kratzer (1977, 1981, 1991), we take the domain of quantification to be the set of best worlds in the modal base according to some ordering source. Let M denote the domain of quantification. To simplify the exposition, we abstract away from the modal base and ordering source, treating M as a simple contextual parameter.

- | | | |
|-----|----|--|
| (5) | a. | $\llbracket \text{must } \phi \rrbracket^M$
is true iff, for <i>every</i> world $w' \in M$, ϕ is true at w' |
| | b. | $\llbracket \text{can } \phi \rrbracket^M$
is true iff, for <i>some</i> world $w' \in M$, ϕ is true at w' |

The quantificational domain M is determined by the flavor of the modal. (See 2. MODAL FLAVOR in this handbook.) Flavor is typically treated as an independent meaning dimension to force. So, in principle, any possible domain M could serve as the domain for either a existential or a universal quantifier.

To summarize, weak modals, or possibility modals, are existential quantifiers, and strong modals, or necessity modals, are universal quantifiers. We sometimes use modal logic notation ($\Diamond$ and $\Box$) as shorthand.

As a reminder, in this chapter we describe work on modal force in the tradition of formal semantics and pragmatics, with an emphasis on the variation observed cross-linguistically in this domain. The typology of modal force has generally otherwise been discussed alongside that of flavor; useful references include [Van der Auwera & Plungian \(1998\)](#), [Nauze \(2008\)](#), [Guo et al. \(2022\)](#), [Uegaki et al. \(2025\)](#). For a more general discussion of modal typology, see 1.5. EMPIRICAL FOUNDATIONS OF MODALITY.

2 Weak necessity

This neat picture, where all modal operators are analyzed as either universal or existential quantifiers over possible worlds, might give one the impression that modal force is an impoverished domain, especially compared to modal flavor. But there is a wrinkle. A certain class of modal operators, the so-called weak necessity modals, appear to be stronger than possibility modals, but weaker than other necessity modals.

2.1 Evidence for weak necessity

Two classic arguments in favor of the claim that certain necessity modals are logically weaker than others come from [von Fintel & Iatridou \(2008\)](#).

2.1.1 Evidence from entailment tests

Under the assumption that necessity modals are universal quantifiers over worlds, and that two modals can quantify over the same set of worlds in a given context (e.g. the set of worlds in which all obligations in that context are satisfied), we predict that certain conjunctions of modal sentences should be contradictory. Suppose that $\Box_1$ and $\Box_2$ are deontic modals that quantify over the same set of worlds. Under this assumption, the prediction is that conjunctions of the form $\Box_1\phi \wedge \neg\Box_2\phi$ should be contradictory (where both conjuncts share the same prejacent ϕ).

This is exactly what we observe in (6) ([von Fintel & Iatridou 2008](#)).

- (6) Contradictory conjunctions of the form $\Box_1\phi \wedge \neg\Box_2\phi$.
- a. #You must do the dishes, but you don't have to.
 - b. #You have to do the dishes, but you don't have to.
 - c. #You are required to do the dishes, but you don't have to.
 - d. #You have to do the dishes, but you are not required to.

This pattern of judgments indicates that the sentences in (7) are mutually entailing, which is expected if the modal operators *must*, *have to*, and *be required to* quantify over the same set of worlds.¹

- (7)
- a. You must do the dishes.
 - b. You have to do the dishes.
 - c. You are required to do the dishes.

However, not all deontic necessity modals follow this pattern. In English, *ought* ϕ and *should* ϕ can be consistently conjoined with negated modal sentences of the form $\neg\Box\phi$.

- (8)
- Conjunctions that are unexpectedly felicitous.
 - a. You ought to wash the dishes, but you don't have to.
 - b. You should do the dishes, but you don't have to.
 - c. You are supposed to do the dishes, but you don't have to.

This indicates that the following entailments do not hold, contrary to our expectations given the pattern in (6).

- (9)
- a. You ought to wash the dishes. $\nRightarrow$ You have to wash the dishes.
 - b. You should wash the dishes. $\nRightarrow$ You have to wash the dishes.
 - c. You are supposed to wash the dishes. $\nRightarrow$ You have to wash the dishes.

Because these entailments do not go through, the English modal operators *ought to*, *should*, and *be supposed to* each appear logically weaker than *have to*.

2.1.2 Evidence from triviality judgments

Certain conjunctions of quantified sentences are infelicitous, arguably because they are trivial: The second conjunct contributes nothing given the first conjunct.

- (10) #Every dog barked, and (in fact) all of them did.

If necessity modals are universal quantifiers over worlds, we should therefore expect that sentences of the form $\Box_1\phi \wedge \Box_2\phi$ should be trivial, and thus infelicitous (again, where $\Box_1$ and $\Box_2$ are universal modals quantifying over the same domain).

- (11)
- a. #You have to wash the dishes, and (in fact), you must.
 - b. #You must wash the dishes, and (in fact) you have to.

¹ These entailment tests could be taken as evidence that this set of deontic modals tend to share the same quantificational domain unless the context pulls them apart.

Weak necessity modals do not behave this way.

- (12) a. You should wash the dishes, and (in fact) you must.
b. You ought to wash the dishes, and in fact, you have to.

This suggests that, in the conjunctions in (12), the contribution of the second clause is not entailed by the content of the first clause.

- (13) a. You should wash the dishes. $\nRightarrow$ You must wash the dishes.
b. You ought to wash the dishes. $\nRightarrow$ You have to wash the dishes.

This follows immediately if weak necessity modals are logically weaker than strong necessity modals.

2.1.3 A note on modal flavor in these sentences

Many modal operators can be used to express more than one modal flavor. Depending on the discourse context, as well as the tense, aspect, or mood morphology of the sentence, the same modal can be interpreted with different flavors.²

- (14) Sue should be at work.
Context 1: We are wondering where Sue is. (epistemic/circumstantial)
Context 2: We are complaining that Sue played hooky. (deontic)

In the prior sections, each modal is arguably being used deontically, in the sense that the truth conditions depend on duties and obligations. The discussion above assumes that weak necessity modals (*should*, *ought to*) and strong necessity modals (*must*, *have to*) share the same modal flavor in the critical examples. For example, if *should* and *have to* were quantifying over different sets of worlds, the fact that *should* ϕ does not entail *have to* ϕ would not point to a difference in force. We will return to this point in the discussion of von Fintel & Iatridou (2008)'s analysis.

2.1.4 Taking stock

A sentence with a weak necessity modal does not entail its strong necessity counterpart. We have seen two converging sources of evidence for this conclusion, namely judgments about contradiction and judgments about triviality (von Fintel & Iatridou 2008). In principle, there are at least two lessons one could take from this:

- i. Weak and strong necessity modals systematically differ in flavor, and thus quantify over different domains.

² For more on these interactions see 2.5. MODAL FLAVOR-SYNTAX INTERACTIONS and 2.4. THE DIVERSITY CONDITION ACROSS MODAL FLAVORS in this volume.

- ii. There are pairs of weak and strong necessity modals that have the same flavor, but some other semantic distinction prevents them from being equivalent.

By and large, the theoretical approaches to this topic mostly follow the second approach, though some of them can be seen as mixing elements of both. In the next section, we take a look at theories of weak necessity.

2.2 Analyses of weak necessity modals

What is the nature of the weak/strong distinction in necessity modals? Below, we survey some recent proposals that address this question. We aim to present the main ideas behind each account, focusing on the arguments and backgrounding the formal details.

For a more in depth exploration of different analytical approaches to weak necessity, see [Rubinstein \(2020\)](#). The goal here is not to capture every detail and nuance, but rather to give the reader a sense of the variety of perspectives that have been taken, and thereby communicate the richness of modal force as a domain of inquiry.

2.2.1 Weak necessity via quantifier domain restriction

One highly influential theory of weak necessity modals comes from [von Stechow & Iatridou \(2008\)](#). They propose that both weak and strong necessity modals are universal quantifiers over possible worlds. What distinguishes the two is that weak necessity modals quantify over a smaller domain of worlds.

This account is motivated by a cross-linguistic generalization about the morphology of weak necessity modals. In many languages, weak necessity modals are formed by applying counterfactual morphology to a strong necessity modal.

Von Stechow and Iatridou call these languages "transparent OUGHT" languages. For example, in Greek, weak necessity is derived by combining future and past inflection on the modal as in (15).

(15) Greek

- a. #prepi na plinis ta piata ala dhen ise ipexreomenos na to kanis
must NA wash the dishes but NEG are obliged NA it do
'You must wash the dishes but you are not obliged to do it.'
- b. tha eprepe na plinis ta piata ala dhen ise ipexreomenos na to
FUT must.PST NA wash the dishes but NEG are obliged NA it
kanis
do
'You ought to wash the dishes but you are not obliged to do it.'

This combination of future and past marking also appears in the main clause (consequent) of counterfactual conditionals.

In French, a weak necessity modal is built by putting a strong necessity modal into the conditional mood, as in (16).

(16) French

- a. #tu dois faire la vaisselle, mais tu n'es pas obligé
you must do the dishes but you NEG.are NEG obliged
'You must do the dishes but you are not obliged to do them.'
- b. tu devrais faire la vaisselle, mais tu n'es pas obligé
you must.COND do the dishes but you NEG.are NEG obliged
'You ought to do the dishes but you are not obliged to do them.'

In both Greek and French, there is a morphological marking pattern that seems to do double duty in the semantics. The same morphology marks both counterfactuality and weak necessity.

In English, *ought* and *would have to* have different meanings. In transparent OUGHT languages, counterfactual-marked strong necessity modals are actually systematically ambiguous between these two readings:

- i. A weak necessity reading, which has truth conditions comparable to English *ought*.
- ii. A counterfactual reading, which is true iff the prejacent is necessary in a counterfactual scenario.

The second reading is the one that is expected given a transparent compositional semantics. How does the weak necessity reading arise, and how are the two meanings related to each other? Von Stechow and Iatridou consider and reject a scopal theory of the weak necessity reading, where the weak necessity reading arises from a strong necessity modal scoping over the counterfactual marking, which keeps its usual meaning. On the basis of that argument, they conclude that the counterfactual marking is interpreted differently in the weak necessity reading.³ In the weak necessity reading, the counterfactual marking signals the promotion of a secondary ordering source, meaning that an extra ordering source is introduced to narrow the domain restriction of the modal.

Abstracting away from formal details, the key idea is that the transparent OUGHT languages reveal that weak necessity meanings are (at least sometimes) derived from strong necessity meanings. From this, von Stechow and Iatridou conclude that weak necessity arises from an extra component of meaning.

³ One could also try harder to maintain a univocal analysis for the counterfactual morphology. For a proposal along these lines, see Silk (2022).

2.2.2 A route to weak necessity via comparative meaning

Rubinstein (2014) proposes that the meaning of weak necessity modals is fundamentally comparative. Rubinstein proposes a formal theory that makes an explicit split between non-comparative (strong) necessity and comparative (weak) necessity modals. She draws support from the composition of weak necessity meanings in Hebrew via comparative morphology.

- (17) Hebrew Rubinstein 2014 ex. (21a)
 yoter/haxi tov še-hu yitpater, aval hu lo xayav lehitpater
 more/most good that-he will.resign, but he NEG must resign
 ‘It is better/best that he will resign, but he doesn’t have to resign.’

Rubinstein proposes that comparative and non-comparative necessity modals have different semantics. Comparative modals explicitly depend on an ordering source, while non-comparative modals do not. In addition to this difference in assertive meaning, Rubinstein adds a lexical presupposition to comparative modals. A comparative modal presupposes that its ordering source is *negotiable*, meaning that not all discourse participants are committed to the ordering source.

This work departs from Kratzer’s framework by limiting the role of ordering sources to only a strict subset of modal meanings, namely the comparative ones. On Rubinstein’s view, ordering sources are used only to rank live possibilities, not to eliminate lower ranked possibilities from consideration.⁴ While comparative (weak) necessity modals are restricted to the best worlds relative to an ordering source, the domain restriction of strong necessity modals does not involve an ordering source.

2.2.3 Non-quantificational semantics for weak necessity modals

Agha & Jeretič (2022) propose that weak necessity modals are non-quantificational. Weak necessity modals are scopeless with respect to negation, which can be seen as analogous to homogeneity effects observed with plural reference.⁵

Prior work, including especially Rubinstein (2014), had addressed this property of weak necessity modals, treating it as a kind of neg-raising. There are also strong necessity modals, like English *must*, that also show a neg-raising pattern, while other strong necessity modals like *have to* are not neg-raisers.

- (18) a. He should not go. ✓ □ > ¬ ; *¬ > □
 b. He must not go. ✓ □ > ¬ ; *¬ > □

⁴ Compare to von Stechow & Iatridou (2008), where an extra ordering source is used to restrict the domain of quantification.

⁵ For more on homogeneity effects, see Löbner (1985), Gajewski (2005), Križ (2016) and references therein.

- c. He doesn't have to go. $*\Box > \neg; \checkmark \neg > \Box$

However, Agha and Jeretič observe that weak necessity modals come apart from other neg-raising modals when embedded under a higher clause negation.

- (19) a. Ann doesn't think Bill should go. $\checkmark \Box > \neg; * \neg > \Box$
 b. Ann doesn't think Bill must go. $*\Box > \neg; \checkmark \neg > \Box$
 c. Ann doesn't think Bill has to go. $*\Box > \neg; \checkmark \neg > \Box$
- (20) a. It's not the case that Bill should go. $\checkmark \Box > \neg; * \neg > \Box$
 b. It's not the case Bill must go. $*\Box > \neg; \checkmark \neg > \Box$
 c. It's not the case that Bill has to go. $*\Box > \neg; \checkmark \neg > \Box$

On their view, this distinction reveals that the theory must make a cut between weak necessity modals like *should*, which never take scope below negation, and neg-raising (strong) necessity modals like *must*, which can take scope below negation when the negation is in a higher clause.

The proposal is that weak necessity modals are not quantifiers, but rather plural predication structures.⁶ On this theory, weak necessity modals are not quantifiers at all, and strictly speaking, they have no scope. The resulting reading merely resembles a wide scope reading for the modal, similar to plural predication structures under negation. For example, (21) means that Ann ate none of the cookies (and not 'Ann didn't eat all of the cookies', which would be expected if the definite plural were a universal quantifier).

- (21) Ann didn't eat the cookies.

Should ϕ is decomposed into (i) a referring expression (the modal, e.g. *should*) that denotes a plural world pronoun, and (ii) the prejacent proposition, which is directly applied to the worlds denoted by the modal pronoun.

- (22) It's not the case that Bill should go to the party.
Paraphrase: It's not the case that the best worlds are worlds where Bill goes to the party.

2.2.4 Other routes to weak necessity meanings

Vander Klok & Hohaus (2020) discuss a distinct compositional route to weak necessity that does not involve counterfactual morphology. In Javanese, a strong

⁶ The idea that natural language makes use of pluralities of possible worlds has been challenged on empirical grounds. See Schmitt (2023).

necessity modal can be suffixed with a morpheme *-ne* to derive a weak necessity meaning.

- (23) wong wong jawa kudu-ne iso ngomong kromo, terus
 person person java ROOT.NEC-NE CIRC.POS AV.talk high.speech, then
 anak-e rojo yo kudu iso
 child-DEF king PRT.YES ROOT.NEC CIRC.POS
 ‘Javanese people ought to be able to speak Krama, and the Sultan’s son has
 to be able to.’

Crucially, the morpheme *-ne* is never used mark counterfactual mood in other contexts, and cannot combine with possibility modals or other verbs (in contrast with counterfactual marking). Interestingly, the *-ne* suffix is homophonous with a nominal suffix that marks definiteness.⁷ This suggests that, in addition to lexicalized weak necessity modals (like *ought*), and modals derived by adding counterfactual marking to strong necessity modals, there is at least one more derivational path to weak necessity.

2.3 Variable force between weak and strong necessity

We have seen several examples of languages where weak necessity modals are built from strong necessity modals by the addition of overt morphology. There are also languages where the same modal expression can express both kinds of meanings. Weingartz & Hohaus (2024) describe such modals in Afrikaans and Samoan (see also Hohaus et al. (2023)).

In Afrikaans, the modal *moet* can be translated into English as either *must* or *should*.

- (24) *In England, the law states that when you ride a motorbike:*
 Jy moet ’n helm dra.
 you NEC a helmet wear
 You must wear a helmet.
- (25) Jy moet nou daardie oproep maak.
 you NEC now that call make
 ‘You should now make that call.’

Crucially, *moet* can be used in a way that carries the empirical signature of weak necessity. Observe that in (26), the speaker is able to consistently assert both (a) and (b).

⁷ This fact could be taken as indirect evidence in favor of a non-quantificational analysis of Javanese weak necessity modals, though Vander Klok & Hohaus (2020) propose a quantificational analysis.

(26) *It's your last day at work before your leave, and there's a lot left to do. You catch sight of your calendar and realise the summer party is this afternoon. Talking to your colleague:*

- a. Ek moet nog na die partyjie toe gaan!
I NEC still to the party to go
'I still should/have to go to the party!'
- b. Maar eintlik hoef ek nie te gaan nie.
but actually NEC I NEG to go NEG
'But actually, I don't have to.'

Weingartz & Hohaus (2024) show that the same pattern holds for Samoan *tatau*.

(27) E tatau ona 'e ta'u=avanoa i tagata savavali pe'a liliu i le itu
TAM NEC that you make=space to person walk.PL if turn to the side
agavale a'o mumū mai le moli lanumumū.
left while red from the light red
'You must give way to pedestrians when turning left at a red light.'

(28) Same context as (26).

- a. E tatau ona ou alu i le pātī.
TAM NEC that I go to the part
I should go to the party.
- b. Ae e lē tatau iā te a'u.
but TAM not NEC to PRN.1.SG
But I don't have to.

Weingartz & Hohaus (2024) choose to analyze Afrikaans *moet* and Samoan *tatau* as weak necessity modals. Assuming the framework of von Fintel & Iatridou (2008), they argue that these modals take a secondary ordering source, and thus have the same basic denotation as the weak necessity modals that have been described in English, Dutch, and Romance. On this view, the difference between the Afrikaans and Samoan modals and the other cases is that, in Afrikaans and Samoan, the secondary ordering source can be empty. The possibility of leaving the secondary ordering source empty delivers an optional strong necessity interpretation. It remains an open question whether any other weak necessity modals optionally have strong necessity readings, and whether the content of the ordering source should serve as the locus of variation for all such cases.

2.4 Weak necessity and weak possibility

To our knowledge, the notion of a possibility counterpart to weak necessity has not received as much attention. Counterfactually-marked possibility modals in languages can be thought of as weak possibility modals, such as English *could* or French *pourrait*.

Outside of the languages where modals can be counterfactual-marked, it is not straightforward to describe what would count as a weak possibility modal. For example, the Javanese strategy for deriving weak necessity modals doesn't apply to possibility modals.

There is an intuition that the meanings of *could* or *pourrait* are somehow weaker or more tentative than their non-marked counterparts *can* and *peut*. Domain restriction accounts of weak necessity do not help to account for this intuition. A secondary ordering source, as proposed by von Fintel & Iatridou (2008), should restrict the modal domain and make the modal claim stronger. Likewise, the plural semantics for weak necessity proposed by Agha & Jeretič (2022) does not have an obvious parallel for existential modals.

To account for the intuition, there may be a way to pursue a comparative possibility approach (e.g. Lassiter 2011) or a modal past account of the counterfactual (e.g. Silk 2022).

2.5 Summary and open questions

If all necessity modals are universal quantifiers over worlds, then the cut between strong and weak necessity is not, strictly speaking, a difference in quantificational force. For example, weak necessity may be derived by additional domain restriction, as in von Fintel & Iatridou's (2008) account. On the other hand, a non-quantificational semantics for weak necessity modals may have some advantages when it comes to the neg-raising facts.

In language after language, it appears that the tense-aspect-mood (TAM) marking found in counterfactual conditionals (also known as X-marking) is also found on weak necessity modals (von Fintel & Iatridou 2023). Thus, in these languages, the problem of how to analyze weak necessity is likely connected to the question of how to derive the meanings of counterfactual conditionals.

Finally, it should be emphasized that the modals that seem to convey weak necessity are morphosyntactically heterogeneous. Some are morphologically simplex, like English *ought* and *should*.⁸ Others, such as the weak necessity modals in transparent OUGHT languages, are derived by adding TAM marking to strong modals. And

⁸ English *should* is arguably simplex in the synchronic grammar of English, but etymologically it descends from a past-marked form of the strong necessity modal *shall*.

others are derived in different ways, such as via comparative syntax (Hebrew) or definite marking (Javanese). Given this heterogeneity, there is no guarantee that all such meanings ought to receive a unified semantic analysis.

With this in mind, here are some open research questions concerning the force of weak necessity modals:

- Do weak necessity meanings express quantification over possible worlds (as opposed to, e.g. plural predication)?
- In transparent OUGHT languages, what is the correct analysis of the distinctive TAM marking (also known as X-marking) observed in counterfactual conditionals and weak necessity modals?
- We observe both TAM-based strategies (as in Greek) and non TAM-based strategies (as in Hebrew and Javanese) for building weak necessity meanings. Should these receive a unified analysis?
- Do ‘weak possibility’ modals exist? What are their properties?

Finally, we want to emphasize that weak necessity is an extremely rich topic that is not exhausted by questions about modal force. We encourage the reader to consult [Rubinstein \(2020\)](#) for a broad overview of weak necessity modals, including their role in modal concord phenomena, their use in discourse, and their connection to gradable modal adjectives.

3 Variable force modals

In recent years, an increasing number of *variable force* modals have been discussed in the literature.⁹ Descriptively, variable force modals seem to express either possibility and necessity readings, depending on their context of use and/or linguistic environment. As we will see, work on this topic has tended not to treat variable force modals as being lexically ambiguous between existential and universal quantifiers. The assumption is that each modal has a single basic lexical meaning with a fixed quantificational force. The other reading is derived from an interaction between the basic meaning and certain context-dependent mechanisms.

Broadly, two kinds of context dependence have been described in the literature on variable force. For some variable force modals, their force seems to only depend on the discourse context in which they are uttered (Section 3.1). Following the proposal in [Rullmann et al. \(2008\)](#)’s seminal paper, these modals are taken to be

⁹ See [Newkirk \(2022a\)](#) for an overview of this literature, which the content of this section is partially modeled after.

universal quantifiers that can quantify over domains of various sizes, yielding weaker interpretations when the domain shrinks, simulating a possibility reading.¹⁰

On the other hand, there are modals whose variable force depends on the polarity of their environment (Section 3.2). Following Deal (2011), these are treated as existential quantifiers whose universal-like reading obtains due to the lack of a scalar implicature, as a simple entailment of the existential quantifier reading. The universal reading disappears in downward entailing contexts, where it is no longer an entailment. Subsequent literature on polarity sensitive modals builds on this analysis, proposing that the universal inference is grammatically encoded using a covert exhaustivity operator (Jeretič 2021a, Newkirk 2022a).

Given the role of exhaustification in polarity sensitive variable force modals, it is natural to wonder whether overt exhaustifiers (like *only* and *even*) can combine with possibility modals to yield necessity readings. In Section 3.3, we discuss data involving possibility modals under exhaustifiers, and see what lessons can be drawn about variable force modals. We suggest that in most cases, the data with *only* look systematically different from variable force modality, but those with *even* might be more closely aligned. Finally, in Section 3.4, we present another phenomenon that falls under variable force modality. Yanovich (2016) describes a pattern called “collapse variable force,” where some modals can only be used in contexts in which possibility and necessity meanings are equivalent.

From this discussion we exclude weak necessity modals (Section 2), which could themselves be argued to constitute cases of variable force modality (see Staniszewski 2019). We also refer the reader back to section 2.3 for examples of modals whose force varies between weak and strong necessity. Finally, we also observe variable force with covert modality, which we leave to Section 4.

3.1 Discourse sensitive variable force modals

In this section we discuss modals whose force interpretation seems to depend on the discourse context only. The key case studies mentioned here come from the languages St’át’imcets (Rullmann et al. 2008), Washo (Bochnak 2015a,b), Gitksan (Peterson 2010, Matthewson 2013), and Nsyilxcen (Menzies 2013). In each case, polarity sensitivity is either shown to be absent, difficult to observe for syntactic reasons, or not discussed. Because these variable force modals are thought to be discourse sensitive, but not polarity sensitive, the formal analyses given in the works discussed tend to adopt similar assumptions to Rullmann et al. (2008)’s influential analysis of St’át’imcets.

¹⁰ Note that this idea is in the spirit of von Stechow & Iatridou (2008)’s account of weak necessity modals, in which a secondary ordering source functions to shrink the quantificational domain of the modal.

St'át'imcets. Rullmann et al. (2008) constitutes the first explicit description and formal analysis of variable force. It is found across items in the modal system of St'át'imcets, which is comprised of modal clitics, each encoding a particular modal flavor, but having variable force. Necessity readings are described to be the default interpretation, but possibility readings are also available. The three main modals with such variable force are epistemic *k'a*, future *kelh* and deontic/irrealis *ka*. We list examples for possibility and necessity uses for each of these items below:

(29) Inferential *k'a*:

- a. **k'a** lh-zúqw-as tu7 ni7 na núkw-a
 INFER COMP-die-3CONJ then DEMON DET other-DET
 qelhmín smúlhats **k'a** lh-mím'c-as tu7 nka7
 old.person woman INFER COMP-move-3CONJ then where
 'Maybe the other old woman died or maybe she moved somewhere.'
- b. *Jim Hoffmann thought he saw a sasquatch and came running back with huge terrified eyes.*
 ka-q'us-tum'-á **k'a** wi7
 CIRC-frighten(CAUS)-PASS-CIRC INFER EMPH
 'It really must have frightened him!'

(30) Future *kelh*:

- a. *You are robbing a house with your friend and you tell him 'Hurry, a policeman might come!' It's not that you know there's a policeman or you see a policeman coming, you are just nervous that one might come.*
 xwem-en-tsút malh! ts7as **kelh** ku plísmen!
 fast-DIR-REFL ADHORT come FUT DET policeman
 'Hurry! A policeman might come!'
- b. o, xílh-ts-kan **kelh** áti7, nilh t'u7
 oh do-CAUS-1SG.SUBJ FUT DEIC FOC just
 s-lh-nás-acw í7wa7
 NOM-COMP-go-2SG.CONJ accompany
 'Oh, I'll do it, if you come along.'

(31) Deontic *ka*:

- a. *You are going for a job interview and the receptionist outside the office tells you that you can leave your bag there, but you can also take it with you when you go in.*

lhwal-en-lhkácw **ka** lts7a ta wa7
 leave-DIR-2SG.SUBJ DEON DEIC DET IMPF
 s-zácen-su; kwán-lhkacw
 NOM-carry-2SG.POSS take(DIR)-2SG.SUBJ
 lh-xát'-min'-acw
 COMP-want-RED-2SG.CONJ

‘You can leave your stuff here; take it if you want to.’

- b. *I don't remember if we ate the rabbits or not.*

t'u7 wa7 **ka** n-scwákwekw-a ts'áqw-an'-em nilh
 but IMPF DEON 1SG.POSS-heart-DET eat-DIR-1PL.ERG FOC
 (t) s-pápt-s-a wa7 tecwecw-wít lh-as kwís-alt
 (DET) NOM-always IMPF increase-3PL COMP-3CONJ fall-child
 i sqweyíts-a
 DET.PL rabbit-DET

‘But I think we had to eat them because they were always having babies.’

Rullmann et al. (2008) analyze variable modals in St'át'imcets as akin to specific indefinites in English, where a choice function variable selects a subset of the set of possible worlds in the modal base. A distributive operator then universally quantifies over the chosen subset of worlds. On this analysis, the force of the modal is always universal; it does not vary. The apparent variability arises because the choice function that determines the domain of quantification can select a larger or smaller subset of accessible worlds. Picking a smaller subset results in a weaker meaning that resembles that of an existential modal statement.¹¹

- (32) a. $\llbracket \text{modal} \rrbracket^{c,w}$ is defined iff c provides a modal base B (inferential, future or deontic/irrealis, depending on the lexical item).
 b. $\llbracket \text{modal} \rrbracket^{c,w} = \lambda f_{\langle st, st \rangle} . \lambda p_{\langle s, st \rangle} . \forall w' [w' \in f(B(w))]$

This analysis captures the discourse sensitivity of the St'át'imcets modals.¹²

Washo. Another case of discourse-sensitive variable force modality is the Washo modal verb *-e?*, as described by Bochnak (2015a,b). Washo *-e?* appears to have three

¹¹ Rullmann et al. (2008) apply the same analysis to other modals, namely the indirect evidential *-an'* and reportative *ku7*. See also Davis et al. (2009), by the same authors, on the circumstantial modal circumfix *ka-...-a*.

¹² One might wonder if these modals are also polarity sensitive. However, these modals are second position clitics, and they seem to not be embeddable under negation and other downward entailing environments, which makes it difficult to tell.

forces, strong necessity, weak necessity and possibility. Its flavor can be deontic, metaphysical, epistemic, bouletic, generic, and pure circumstantial.¹³

- (33) *Context: You are planning to drive over the mountains. It has started to snow, and you know that whenever it snows, the road over the mountains is closed.*
 déʔeš-ʔaŋaw-i-š yéweš gum-beyéč'ig-i-gi k'-éʔ-i
 snow-good-IND-SR road REFL-close-IND-REL 3-MOD-IND
 'It's snowing a lot, so the road must be closed.' EPISTEMIC NECESSITY
- (34) *Context: You hear a knock at the door. You can't see through the window who it is, and you're not expecting anyone, but you can make out that the person looks about the same height as Beverly.*
 bévali k'-éʔ-hel-i-gi k'-éʔ-i
 Beverly 3-be-SUBJ-IND-REL 3-MOD-IND
 'It might be Beverly.' EPISTEMIC POSSIBILITY
- (35) *Context: A friend comes to visit, and brings her dog along. You don't want the dog to come inside.*
 súku baŋáya ʔ-éʔ-i-š-gi k'-éʔ-i
 dog outside 3-be-IND-SR-REL 3-MOD-IND
 'The dog has to stay outside.' DEONTIC NECESSITY
- (36) *Context: Mary's friends come over to ask her to come play. She isn't allowed, because she hasn't finished all her chores. Later, her friends return.*
 wádiŋ hé:š ʔum-p'àyt'iʔ-giš-uweʔ k'-éʔ-i
 now Q 2-play-along-hence 3-MOD-IND
 'Now are you allowed to come play?' DEONTIC POSSIBILITY

Bochnak tests for polarity sensitivity by embedding *-eʔ* inside the antecedent of a conditional. He observes that both possibility and necessity readings are still available in this environment, and concludes that *-eʔ* is polarity-insensitive.¹⁴ The resulting analysis resembles [Rullmann et al. \(2008\)](#)'s discourse sensitive theory, but with the modal flavor left unspecified.

13 Notably, Washo *-eʔ* displays both force and flavor variability. Washo thus presents a counterexample to [Nauze \(2008\)](#)'s conjecture: "Modal elements can only have more than one meaning along a unique axis of the semantic space: they either vary on the horizontal axis and thus are polyfunctional in the original sense of expressing different types of modality or they vary on the vertical axis and can express possibility and necessity, but they cannot vary on both axes." See 1.5. EMPIRICAL FOUNDATIONS OF MODALITY in this volume for a more extensive discussion about Nauze's conjecture.

14 Due to syntactic restrictions, *-eʔ* cannot be embedded under negation. In Section 3.2, we present cases in which embeddings under conditional antecedents are not enough to show a lack of polarity sensitivity, and it is crucial to test clausemate negation.

Gitksan. In Gitksan, two epistemic modals are claimed to have variable force (Peterson 2010, Matthewson 2013): inferential =*ima* and reportative =*kat*.

- (37) *Context: The speaker's father was away frequently when she was a child.*
 naa'ay=**ima** 'an yookxw-in-ý
 mother=MOD S.REL eat-CAUS=1SG
 'It must've been mother who fed me.'
- (38) *Context: Alvin's wife asks when Alvin is getting back from shopping in Smithers.*
 íagý, witxw=**ima** níit silkwsax
 forget-1sg arrive=mod 3sg at.noon
 'I forget, he might arrive around noon.'
- (39) m̓ats-i-(t)=**kat**=hl ha-n̓ii-kuyǰax 'a=hl lo'op
 hit-TR-3=REP=CND INSTR-in-light LOC=CND rock
 a. 'I hear he hit the window with a rock (and broke it).' (*Context: report of reliable adult*)
 b. 'Apparently, he hit the window with a rock.' (*Context: report of child trying to escape blame*)

On both readings, the speaker commits to the possibility of the prejacent, and the evidence source is a presupposition. In fact, Peterson argues for a default preference for possibility readings of these two modals, based on consultants' translations out of the blue. This leads him to choose an analysis different from Rullmann et al. (2008), who describe St'át'imcets modals to have a default necessity interpretation. He proposes that Gitksan modals' variable force is due to the ability to switch ordering sources (following Izvorski 1997).

- (40) a. $\llbracket =ima \rrbracket^{c,w}$ is defined iff c provides a modal base B such that for all worlds $w' \in B(w)$, the inferential evidence in w holds in w' .
 b. $\llbracket =ima \rrbracket^{c,w} = \lambda p. \exists w' [w' \in O_{g(w)}(B(w)) \wedge p(w') = 1]$

Peterson argues that making the ordering source more specific allows for a strengthening of the modal claim. While technically true, it is unclear how strengthening via specificity can result in a *must*-like reading.

Nsyilxcen. Menzies (2013) describes the modal system of Nsyilxcen, which consists of two epistemic modals, *mat* and *cmay*, in addition to a reportative and bouletic modal, not discussed here. Out of the two epistemic modals, *mat* can express both possibility and necessity readings, while *cmay* only expresses possibility. They both are felicitous in inference contexts.

Below is a context that targets the possibility modal reading: both *mat* and *cmay* are felicitous.

- (41) *Context: You know that Mary loves to go running and often goes on runs randomly. She could also be at the store or at school. I ask you, where is Mary?*
- a. Mary **mat** ac-qíc-əlx
Mary EPIS.MOD CUST-run-LEX
'Mary might be running.'
 - b. Mary **cmay** ac-qíc-əlx
Mary EPIS.MOD CUST-run-LEX
'Mary might be running.'

Below is a context that targets the necessity modal reading. While *mat* is felicitous, *cmay* is not.

- (42) *Context: Mary runs every day to train for a marathon. She usually runs at 6pm on Tuesdays. Today is Tuesday and it's 6pm. I ask you, where is Mary?*
- a. Mary **mat** ac-qíc-əlx
Mary EPIS.MOD CUST-run-LEX
'Mary must have went running.'
 - b. #Mary **cmay** ac-qíc-əlx
Mary EPIS.MOD CUST-run-LEX
'Mary might be running.'

To account for the variable force of *mat*, Menzies follows the analysis by [Rullmann et al. \(2008\)](#). The modal quantifies universally over a proper subset of the modal base, where that subset is selected by a choice function. In contrast, *cmay* is analyzed as a typical possibility modal, with an existential semantics.

3.2 Polarity sensitive variable force modals

In this section we discuss examples of variable force modality that show sensitivity to the polarity of the linguistic environment. We find that the cross-linguistic picture is complex and there is no single pattern to uncover. Like in the last section, we provide the analysis of each case after presenting the empirical profile of the modals in question. The case studies we mention here come from Nez Perce ([Deal 2011](#)), Ecuadorian Siona ([Jeretič 2021a,b](#)), Swedish ([Jeretič 2021a](#)), and Kinande ([Newkirk 2022a,b](#)). We end the section with a synthesis of the empirical profiles of polarity sensitive variable force modals. Readers may find it useful to read the synthesis at the end first, before digging into the details of each case study.

Nez Perce. Deal (2011) describes a deontic modal item *o'qa* in Nez Perce whose force is variable, but only in upward entailing environments (with a possibility reading as a default). In downward entailing ones, only possibility readings are available. In (43), both possibility and necessity readings are available.

- (43) 'inéhne-**no'qa** 'ee kii lepít cíckan
 take-MODAL you DEM two blanket
 a. 'You can take these two blankets.'
 b. 'You should take these two blankets.'

When embedded in downward-entailing contexts, it can only have possibility readings, as shown under negation in (44) and in a conditional antecedent in (45).

- (44) *You are explaining to someone who thinks they have to leave that they are not in fact required to do so. It's not necessary for them to leave.*
 #wéet'u 'ee kiy-ó'qa
 not you go-MODAL
 (intended) 'You don't have to go'.
 (actual, but weird in the context) 'You can't go.'
- (45) *Prompt: If I have to call the doctor, I will.*
 #c'alawi 'a-múu-**no'qa** saykiptaw'atoo-na, kaa 'e-múu-nu'
 if 3OBJ-call-MODAL doctor-OBJ then 3OBJ-call-PR
 (intended) 'If I have to call the doctor, I will'.
 (actual, but not the prompt) 'If I can call the doctor, I will.'

Deal (2011) analyzes *o'qa* as a possibility modal. There is no necessity modal in the language that could act as a scalar alternative, and therefore statements of the form *o'qa p* do not generate scalar implicatures of the form *not have to p*. As a result, in unembedded contexts, *o'qa* can be used to express necessity meanings, as a special case of the possibility reading. In downward-entailing contexts, however, the necessity reading is globally weaker than the possibility reading, therefore not a special case of it. As a result, *o'qa* cannot have a necessity reading in a downward-entailing context.

Deal's analysis of Nez Perce has been highly influential in the literature on variable force modality. As we will see, the intuition that variable force modals are possibility modals that lack a strong scalemate lies behind much subsequent work on the topic.

Ecuadorian Siona. In Ecuadorian Siona, there is a modal *ba'iji* that also displays polarity sensitivity, but with a slightly more complex distribution than Nez Perce

o'qa (Jeretič 2021a,b). In unembedded environments, Siona *ba'iji* appears to only have necessity readings; when negated, it is unambiguously possibility; in any other other non-upward-entailing environment, both forces are possible.

Below is data from Siona supporting this generalization. Unembedded *ba'iji*, when uttered out of the blue, displays necessity interpretations as in (46), but not possibility interpretations, as attempted in (47).

- (46) *Context: San Pablo is on the other side of the river. A asks a stranger, B, how to get there. B answers:*
 Tsiaya-jã'ã je'e-ñe **ba-'i-ji**.
 river-PATH cross-INF be-ASRT
 'One has to cross the river.'
- (47) *Context: we're going back to the village. I see a path, I wonder where it goes.*
 #Iye ma'a-jã'ã ti'añe **ba-'i-ji** colegio-na... Jare iye ma'aja
 there path-PATH get be-3s school-GOAL... that this path-OBL
 cuine-je ba-'i-ji ti'añe gaaëyohueña.
 also-ADD be-ipf-3s get soccer.field
 #'On this path you have to get to the school... And on this same path you have to also get to the soccer field.'

A *ba'iji* utterance can be negated, by using a negated version of the copula *beoji*. The only reading available is a prohibition reading.

- (48) Sai-ye **beo-ji**.
 go-INF NEG.be-3s
 'We mustn't go.'
Impossible reading: 'We don't have to go.'

In other downward-entailing environments such as conditional antecedents, both forces are available: (49-a) is a context that favors a possibility reading, while (49-b) favors a necessity reading; *ba'iji* is possible in both.

- (49) *Context: I am waiting to see if there is going to be a spot for me in the boat. My friend asks me if I want to go.*
- a. – Yë-yë. Sai-ye **ba-'i-to**, sa-si-'i.
 want-1S go-INF be-IPF-COND go-FUT-OTH
 'Yes. If I can (#must) go, I will go.'
- b. – Coe-yë. Sai-ye **ba-'i-to**, sa-si-'i.
 neg.want-1S go-INF be-IPF-COND go-FUT-OTH
 'No. But if I must (#can) go, I will go.'

Jeretič (2021a) analyzes modals with this type of distribution as possibility modals that trigger a scaleless implicature. A scaleless implicature is a universal inference that results from a weak scalar item whose strong scalemate is absent from the lexicon.¹⁵ This type of implicature can be derived in a grammatical framework for scalar implicatures (Fox 2007, Chierchia et al. 2012). While the mechanism for obtaining the necessity reading from the existential one is slightly different in Nez Perce and in Siona, the two rely on the lack of a necessity scalemate.¹⁶

Formally, the difference between the analyses for Nez Perce and Siona is that in Siona, *ba'iji* triggers subdomain alternatives while *o'qa* does not. Subdomain alternatives are alternatives of a quantificational sentence obtained by replacing the domain of the quantifier (here, the modal domain) by each of its subsets. An exhaustification operator (EXH) (as proposed by Fox (2007) or Bar-Lev & Fox (2020)) then applies in the syntax, mimicking the Gricean pragmatic mechanism of exclusion of alternatives utterances. When it operates over subdomain alternatives of an existential quantifier, it forces a universal reading.

We give a simplified formal illustration of this for *ba'iji*. The meaning of a *ba'iji* sentence is an existential quantifier over a domain of worlds M , as shown in (50-a); its alternative set is defined in (50-b), and consists of subdomain alternatives, but no scalar alternative. An exhaustifier EXH operates over these alternatives, and yields a necessity meaning, as shown in (50-c). See Jeretič (2021a,b) for details.

- (50) a. $\llbracket \text{ba'iji}_M p \rrbracket = \exists w \in M. p(w)$
 b. $\text{Alt}(\text{ba'iji}_M p) = \{ \exists w \in M'. p(w) \mid M' \subseteq M \}$
 c. $\llbracket \text{EXH ba'iji}_M p \rrbracket \equiv \forall w \in M. p(w)$

General constraints on the distribution of EXH account for the interpretations of *ba'iji*: EXH applies at clause boundaries, and it is obligatory whenever it strengthens the utterance (except under a specific type of QUD), and optional otherwise. As a result: in upward-entailing contexts, EXH strengthens the utterance and is thus obligatory; elsewhere, like in conditional antecedents, it is optional. There is no necessity reading under negation because of the lack of clause boundary between negation and the modal.

¹⁵ Here we discuss cases of possibility modals strengthened to necessity meanings. Jeretič (2021a) argues that scaleless implicatures can also be triggered by negated universal modals like negated English *must*, where a strong reading equivalent to $\Box \neg$ is obtained from an underlying weak reading $\neg \Box$. This analysis is proposed as an alternative to the PPI analysis of these modals, which is discussed elsewhere in this volume. See 6.1. MODALS, NEGATION, CONCORD, AND POLARITY.

¹⁶ Refer to Jeretič (2021a) for a discussion of what counts as an appropriate scalemate to derive scalar implicatures.

With some subtle differences, Slovenian *moči* and French *avoir à* are claimed to have a similar distribution to Siona *ba'iji*,¹⁷ and receive the same analysis. In Jeretič (2021a), these three cases are part of a class of variable force modals that trigger ‘obligatory scaleless implicatures’, and they are characterized by having a fixed universal force in default unembedded contexts, and revealing their underlying possibility reading only in particular linguistic or discourse contexts.

Swedish. Another class of variable force modals identified in Jeretič (2021a) are those triggering ‘optional scaleless implicatures’, which have essentially the same distribution of forces as the above class except that both possibility and necessity readings are available in default unembedded contexts. One example of such a modal is *får* in Swedish. It is ambiguous between possibility and necessity readings, everywhere except under negation. Below is the data supporting this generalization. Unembedded, *får* is ambiguous between a possibility and necessity modal, as shown in examples (51) and (52).

- (51) Alice **får** gå ut, men hon **får** också stanna.
 Alice fa go out but she fa also stay
 ‘Alice is allowed to go out, but she is also allowed to stay.’
- (52) *I’m telling a story in which Isac illegally parked the car, and the police caught him.*
 Isac **får** betala en bot.
 Isac MOD pay a fine
 ‘Isac has to pay a fine.’

Under negation, only the permission reading is available, resulting in a prohibition.

- (53) *Peter is a prisoner.*
 Peter **får** inte lämna fängelset.
 Peter MOD not leave prison
 ‘Peter is not allowed to leave the prison.’

In contrast, a context which makes only the negated necessity reading available is incompatible with a felicitous use of negated *får*.

- (54) *The rules in this building are pretty relaxed. The doorman says:*
 #Du **får** inte visa bevis på vaccination för att komma in.
 you MOD not show proof of vaccination before to come in

¹⁷ With a stronger tendency towards necessity readings than Siona. Slovenian and French show an additional property of the distribution of possibility modals: they are forced under perfective readings.

int. ‘You don’t have to show a proof of vaccination to enter.’

Embedded in non-UE contexts other than negation, it is again ambiguous. Below are examples in conditional antecedents.

- (55) *Lucas loves ice cream, but only eats it when his mom gives him permission.*
 Om Lucas **får** äta glass blir han glad.
 if Lucas MOD eat ice.cream be.fut he happy
 ‘If Lucas is allowed to have ice cream, he will be happy.’
- (56) *Maria took the train without paying and is worried about getting caught.*
 Om Maria **får** betala en böter blir hon olycklig.
 if Maria MOD pay a fine be.fut she unhappy
 ‘If Maria has to pay a fine, she will be unhappy.’

Jeretič (2021a) analyzes *får* similarly to the modals of the Siona *ba’iji* type. The availability of the possibility reading in unembedded contexts arises from deriving a ‘not have to’ scalar implicature; this can occur due to the presence of a necessity scalemate *behovå* in Swedish. The necessity reading of *får* is a scaleless implicature that can occur if the scalar alternative *behovå* is pruned from the set of alternatives. Under negation, no implicature is derived and only a possibility reading is available, just like for Siona *ba’iji*. In the other environments, the exhaustifier is optionally available to derive a scalar or scaleless implicature.

The distribution of the Swedish modal furthermore differs from the Nez Perce *o’qa* from its behavior in non-UE environments, where *o’qa* only has a possibility reading. Jeretič argues that the analysis for *o’qa* provided by Deal (2011) can co-exist with the scaleless implicature analysis, if we assume that Nez Perce *o’qa* projects no subdomain alternatives, which accounts for the lack of strengthening in downward entailing environments.

Kinande. Another example of a variable force modal is Kinande *anga*, described and analyzed by Newkirk (2022a,b): it allows for possibility and weak necessity readings across flavors (except in the teleological flavor, where only possibility readings are available). Under clausemate negation, it can only have a possibility reading, while it is ambiguous in all other environments (including downward entailing environments such as antecedents of conditionals, restrictors of universal quantifier, and under extra-clausal negation). Strong necessity readings are never available for *anga*.

Below is the data provided by Newkirk (2022a) to support this generalization.

- (57) **Epistemic possibility:** Kábunga is not a consistent student. The professor never knows if he will come to class or not. Today, it is time to start class

and he has not shown up yet.

Epistemic weak necessity: Kábunga is not a consistent student, and often skips class on Wednesdays, but he is usually there on Mondays. Today is Monday, and so the professor isn't entirely sure whether Kábunga will show up, but since it's Monday and Kábunga is relatively consistent on Mondays, he thinks Kábunga will probably be there.

Kabunga a-**anga**-na-sy-a oko kalhasi ko munabwire

Kabunga SM.c1-MOD-TM-come-FV c17 today

“Kabunga might/should be coming to class today”

- (58) **Negated deontic possibility:** A player has been hurt in a football match, and the ref is speaking to them, and saying:

nga-oko reglema yi-ka-bug-a,

COMP-COMP c9.rules SM.c9-TM-say-FV,

si-u-**anga**-sat-a

NEG-SM.2sg-MOD-dance-FV

“According to the rules, you can't play.”

- (59) **Negated deontic strong necessity (unavailable):** Kabunga often plays football with his friends, but today he is tired and doesn't want to play. One of his friends tells him that it's ok, he doesn't have to play today if he doesn't want to:

#sí-u-**anga**-sat-a

NEG-SM.2sg-MOD-dance-FV

Intended: “you don't have to play”

- (60) **Deontic strong necessity (unavailable):** In Uganda, the law states that when you ride a motor bike...

#u-**anga**-na-ambal-a e-sapu

2sg-MOD-TM-wear-FV c9/3-helmet

Intended: “You must wear a helmet”

Newkirk analyzes this intricate pattern as the coexistence of a scalar implicature and a scaleless implicature: *anga* can strengthen via exhaustification from its underlying possibility semantics to a weak necessity reading, but not to a strong necessity reading, because of the presence in the language of the strong necessity modal *paswa*. Its weak necessity semantics originates from the domain of the modal, obtained from restricting the modal base with two ordering sources, as proposed in von Fintel & Iatridou (2008):

$$(61) \quad \llbracket \text{anga} \rrbracket^w(p) = \exists w' [w' \in OS_2(OS_1(\bigcap(MB(w)))) \wedge p(w')]$$

Newkirk claims that as a possibility modal, this domain restriction has no noticeable effect. However, when the interpretation is strengthened via scaleless implicature to a universal meaning, the result is a weak necessity modal meaning.

Furthermore, the modal can never have a strong reading because of the scalar implicature obtained by negating the strong necessity modal *paswa*, which has the same range of modal flavors as *anga*.

Interim summary. To wrap up this section, we summarize the different types of polarity sensitive patterns and note some key generalizations. The table below gives a partial view of variable force modal typology. In the table, we use the $\square$ and $\diamond$ symbols as labels to describe which readings are possible in which environments. $\square$ signals the availability of a necessity reading and $\diamond$ signals the availability of a possibility reading.

	Pattern 1 Nez Perce <i>o'qa</i>	Pattern 2 Siona <i>ba'iji</i>	Pattern 3 Swedish <i>får</i>	Pattern 4 Kinande <i>anga</i>
unembedded	$\diamond, \square$	$\square$	$\diamond, \square$	$\diamond, \square_w$
clausemate negation	$\diamond$	$\diamond$	$\diamond$	$\diamond$
other DE contexts	$\diamond$	$\diamond, \square$	$\diamond, \square$	$\diamond, \square_w$

The analyses of the polarity-sensitive modals are mostly consistent with one another. The first three patterns are discussed in the modal typology laid out in Jeretič (2021b): *o'qa* projects no alternatives whatsoever, modals in Pattern 2 project sub-domain alternatives but no scalar ones, and those in Pattern 3 project both, but scalar alternatives may be pruned. Furthermore Newkirk (2022a) proposes an expanded typology to include the Kinande modal by adding a secondary ordering source. However the Kinande data questions the difference between Pattern 2 and Pattern 3 being explained by the presence of a scalemate (why can't *anga* have a strong necessity reading like Swedish *får* by pruning the strong necessity alternative?)

This way of looking at the typology reveals another key point. For each variable force modal listed, under clausemate negation, the only available reading is the possibility reading. In other DE environments, such as conditional antecedents, necessity readings are still possible. Recall from Section 3.1 that existing work on discourse sensitive variable force modals does not present evidence of their behavior with clausemate negation. In some cases (e.g. St'át'imcets), this may not be possible to test because of independent restrictions on the distribution of the modal item. But in other cases, the relevant data may not yet have been reported. Thus, the possibility of unifying variable force behaviors under an implicature-based analysis remains open, pending a fuller investigation of the negation facts.

3.3 Building necessity from possibility with an overt exhaustivity marker

Sometimes, possibility modals can have necessity readings when overtly co-occurring with an exhaustive particle like *only* or *even*. Such constructions can therefore be compared with the analyses of the polarity-sensitive variable force modals provided in the previous section: exhaustification would be a route from possibility to necessity meanings, and the exhaustifier could be overt or covert. However, there are reasons to avoid jumping to the conclusion that the overt exhaustifier cases and the covert EXH cases are completely parallel, which we discuss in this section. Indeed, we find that the cases with exclusive *only* are not to be paralleled with scaleless implicatures, but there is more promise with those with inclusive *even*.

3.3.1 Building necessity from possibility and *only*

Leffel (2012) describes that Masalit can express epistemic necessity by combining *only* with an epistemic possibility modal, as shown in (62).

- (62) a. tí màsàrá tú-tì
 3sg Masalit 3sg-may
 He might be Masalit.
 b. tí màsàrá dè tú-tì
 3sg Masalit only 3sg-may
 He must be Masalit.

A similar pattern can be found in Hausa, as described by Grubic & Mucha (2021). A deontic necessity example of this type can be found below (circumstantial necessity and weak necessity is also available), which is argued to involve a covert possibility operator.

- (63) *According to Nigerian law,*
 Sai Audù yà tàfi fuɓɓùnà.
 only audu 3sg.m.sbjv go prison
 Audu must go to jail.

Hungarian exhibits a comparable phenomenon, where the exclusive is not itself overt, but is revealed through a focus structure change that forces the exclusive reading, as described by Kiefer (1986) (and cited by Leffel 2012).

- (64) a. Péter lehet nyelvész.
 Peter be.may linguist
 “Peter may be a linguist.”

- b. Péter [nyelvész]_F lehet.
 Peter linguist be.may
 ‘Peter must be a linguist.’

Mandarin has a construction that involves an exclusivity marker and the particle *de* that is historically related to a possibility modal (Bian & Jeretič in prep.).¹⁸

- (65) zhe zhi duiwu wuzi haojin, zhi de fanhui.
 this CL team supply depleted only DE return
 ‘This team’s supply was depleted, so they had to return.’

However, it is unclear from these examples whether these are cases that directly express necessity modality, or are simply in contexts that entail necessity. We can construct an example in English below that illustrates this problem.

- (66) I can only stay home. $\approx$ I have to stay home.

In this sentence, *only* associates with the focused constituent *stay home*; as a result, alternative possibility claims such as “I can go out shopping”, “I can go for a walk”, are negated, which leaves us with only one live possibility “I can stay home”, yielding the necessity meaning. Such an example is in a sense uninteresting: it is derived fully compositionally, but does not provide a bona fide necessity claim. Indeed, as Leffel (2012) notes, a necessity claim can be compatible with alternative possibilities, as shown in (67-a), but not with this exclusivity construction, as in (67-b).

- (67) a. I must cook food. I may also sweep.
 b. # I may only cook food. I may also sweep.

Furthermore, an exclusive possibility claim does always not force the prejacent to be necessary, if the possibility for the negated prejacent is not an alternative itself, or entailed by the alternatives. Such a case can be constructed in English, showing that *only can* is available, but *must* is not.

- (68) A: What can I do to help you?
 a. B: Really, you can only help me with the dishes. But you don’t have to.
 b. B: # Really, you must help me with the dishes. But you don’t have to.

It is therefore clear in English that this is not a case of necessity modality, but rather there is a necessity modal inference that can be drawn in some contexts. Crucially,

¹⁸ *De* is nowadays restricted to negative environments, and historically related to necessity *děi* (identical in their orthography).

this is different from the cases described in the previous section, of pure necessity modality. Both cases involve an exclusive operator.¹⁹ The difference between them is the focus associate. In English, the focused constituent is a subconstituent of the prejacent of the modal. Therefore, the focus is in the scope of the quantifier. The result is not logically equivalent to a necessity reading. In the EXH-based cases of variable force, the focused element is the domain of the modal, i.e. the restrictor of the quantifier. The result is equivalent to a necessity reading.

What does this suggest about the situation in the cases mentioned above? [Leffel \(2012\)](#) suggests that the data in Masalit is unclear, but supposed to be comparable to English for a sentence such as (67), meaning that the necessity reading is an inference arising in certain contexts. For Hausa, [Grubic & Mucha \(2021\)](#) argue that the exclusive particle *sai* focus-associates with the entire prejacent. This prejacent can be paraphrased as “it is possible that there is a future event of Audu going to prison.” The exclusive particle thus excludes all future possibilities except for the prejacent itself. The resulting reading is inconsistent with the truth of any alternative future possibility, which is not expected from a standard necessity meaning. This analysis therefore predicts that certain followups will be inconsistent in Hausa, along the lines of the test in (67-b). As far as we know, the Hausa data that would be crucial to test this prediction has not been reported.

In Mandarin, the situation is different. The exclusive + *de* construction seems to pattern like a bona fide necessity modal. The following example shows it is incompatible with the possibility that the prejacent be not realized (the replacement of *de* with another possibility modal such as *neng* makes the sentence acceptable).

- (69) #zhe zhi duiwu buji haojin, suoyi tamen **zhi de** zai qianjin
 this CL team supply depleted so they **only de** again march.forward
 ji ge xiaoshi, danshi tamen jue ding liji fanhui.
 several CL hour but they decide immediately return.
intended. ‘This team’s supply was depleted, so they could only further march forward for a few more hours, but they decided to return immediately.’

Bian & Jeretič analyze this construction as a lexicalized necessity modal, which has lost its compositional meaning through lexical reanalysis.

Overall, while the strong readings of some variable force modals are thought to result from covert EXH, there is no clear evidence that those strong readings are equivalent to the meaning of a possibility modal under an exclusive operator.

¹⁹ Covert EXH and overt *only* are thought to differ in whether they presuppose or assert their prejacent, but it is not always possible to detect a difference. In (68), if *only* presupposes its prejacent, the presupposition could be globally accommodated here, in which case the two assertions would be equivalent. Neither [Leffel \(2012\)](#) nor [Grubic & Mucha \(2021\)](#) discuss this point for Masalit or Hausa.

3.3.2 Building necessity from possibility and *even*

In the Mandarin spoken in the 16th centuries, we find necessity readings obtained from a possibility modal *de* combining with particles *hai* and *ye* which had *even*-like readings (Bian & Jeretič *in prep.*). The following examples, all from *Journey to the West* (circa CE 1520-1580), showcase each of these readings.

- (70) **Bare *de*: possibility**
 xupiaopiao dao yu ci di, geng bu shi lujing, jin zen
 light.and.floating arrive at this place further NEG know path now how
de hui xiang?
 DE return home
 ‘We drifted all the way here, light as air, and now we don’t recognize the path — how can we possibly return home?’
- (71) ***hai de*: necessity**
 woliao shouduan bu jian zendi, **hai de** ni xian xia shui.
 1.DU.EXCL skill NEG see how still DE you first down water
 ‘We two do not get good skills, it should be you who get into the water first.’
- (72) ***ye de*: necessity**
 zaojian dianxin **ye de** baishi ge shaobing cai gou.
 morning dim.sum also DE hundred CL pancake then enough
 ‘For breakfast, he needs nearly a hundred pancakes.’

Bian & Jeretič (*in prep.*) analyze these cases as scaleless implicatures with an overt *even*-like exhaustivity marker. There is indeed no obvious alternative compositional way to obtain a necessity reading from the meaning of *even* and that of a possibility modal. This can be seen with the English counterpart *even can*, which cannot entail a necessity reading in any context. This finding is significant for the literature on variable force and scaleless implicatures: an example with an overt exhaustifier has been found, and specifically with an *even*-like exhaustifier.

3.4 Collapse variable force

Alfredian Old English **motan* (the ancestor of *must*) is traditionally described as having variable force between possibility and necessity readings. Below are examples from Oxford English Dictionary to support this claim, cited by Yanovich (2016):

- (73) Possibility-like uses.
 Gif he us geunnan wile, þæt we hine swa godne gretan **moton**.
 if he us grant will that we him so good greet mot.prs.pl

‘If he will grant to us that we moton greet him, the good one.’ (Beowulf 347)

(74) Necessity-like uses.

Londrihtes **mot** þære mægburge monna æghwylc idel
of.landright mot.prs.ind.3sg of.that kin of.men each idle
hweorfan.
wander

‘Every man of that kin mot wander without the rights of the rightful residents.’ (Beowulf 2886)

Yanovich (2016) proposes an analysis of this variable force as a possibility modal with a presupposition of “inevitable actualization”. This presupposition forces **motan* to be used in contexts in which if the prejacent could be actualized, it would. In such contexts, possibility and necessity uses are equivalent. Yanovich calls this effect “collapse variable force.”

He cites other examples that are to be compared with collapse variable force: modals with actuality entailments, ability modals, variable force ‘get’-based modals around the Baltic Sea (including Swedish *får* mentioned earlier in this section). While these are not all the same phenomenon, Yanovich claims that they require similar semantic components. What exactly differs between each of these cases is left open.

Yanovich also argues that Middle English **moten*, the descendant of Old English **motan*, no longer displays the same distribution and seems to be actually ambiguous between a possibility and a necessity modal.

3.5 Summary and open questions

In the literature on variable force modals, several distinct patterns have been described. There are modals whose force is purely discourse sensitive (Section 3.1), modals whose force is polarity sensitive (Section 3.2), possibility modals that associate with overt exclusives, resulting in a necessity-like meaning (Section 3.3), and finally, modals that only appear in contexts where possibility and necessity readings are equivalent (Section 3.4).

There is a split in the literature between approaches to variable force that focus on discourse sensitivity and approaches that focus on polarity sensitivity. However, we still don’t have the full empirical picture when it comes to polarity sensitivity, mostly because the relevant diagnostics are not always applied in a uniform way. The polarity sensitive modals described in Section 3.2 are all shown to only have possibility readings under clausemate negation. There is no data with the discourse sensitive variable force modals in Section 3.1 under clausemate negation, either

because it is not reported, or because the modal under study cannot be embedded under negation. Clausemate negation is a crucial environment to test for polarity sensitivity (as there are some cases like Swedish *får*, which only reveal their polarity sensitivity in this environment). Thus, it remains open whether any of the discourse sensitive variable force modals are also polarity sensitive.

While the literature is divided on how to analyze variable force, it has also converged on the view that each modal has only one basic lexical meaning. In a sense, the common assumption of all these approaches is that modal items do not really have variable force, at least at the level of their lexical meaning.

Our understanding of variable force modals is still developing, and many questions remain open. For example:

- Which variable force modals are polarity sensitive?
- Which cases of polarity sensitive variable force modals can be explained by scaleless implicatures?
- Can we find more cases of necessity meanings obtained from a possibility modal and an *even*-like particle?
- We encounter a variety of analyses of variable force modals in the literature. If each analysis is right for at least some cases, then how does a child learn the meanings of modals in their language?

4 Force phenomena and covert modality

Certain constructions—like conditionals—carry modal meanings, but do not transparently contain modal operators. In formal semantics, these modal meanings are sometimes attributed to covert modal operators, which are taken to be present in the syntactic structure, but phonologically null. Many covert modals appear to have variable force, or weak necessity-like meanings. Here, we briefly discuss some work on the force of covert modals, using English conditionals, generics, imperatives, and infinitival relative clauses as case studies.

4.1 Bare conditionals and generics

In Kratzer's (1986) theory of conditionals, the function of an *if*-clause is to restrict a modal quantifier.

- (75) If Ann comes to the party, Bill **must** come too.
 $\text{MUST}(\{w : \text{Ann comes in } w\})(\{w : \text{Bill comes in } w\})$

On this view, a question arises about how to treat conditional sentences where the main clause lacks an overt modal operator, which are sometimes called *bare conditionals*.

(76) If Ann comes to the party, Bill comes too.

(77) If you work hard, you succeed.

On Kratzer's theory, bare conditionals like those in (76) and (77) do in fact contain covert modal operators, contrary to appearances. This assumption allows us to maintain a uniform treatment of *if*-clauses.

According to this theory, bare conditional sentences are quantificational structures with universal force.

(78) If Ann comes to the party, Bill comes too.

$\Box(\{w : \text{Ann comes in } w\})(\{w : \text{Bill comes in } w\})$

However, the silent universal modal theory encounters difficulties when conditional sentences are embedded. For example, bare conditionals embedded under negation, downward entailing quantifiers, and *only* have meanings that are unexpected on this view (von Fintel 1997, Kratzer 2021, Herburger 2015, 2019, Bassi & Bar-Lev 2018).

Consider the sentences in (79). In each case, the most salient reading is the strong reading that hard work *never* results in success, a reading that strikes one as implausible given world knowledge.

- (79) a. You don't succeed if you work hard.
- b. It's not the case that you succeed if you work hard.
- c. I don't think you succeed if you work hard.
- d. I doubt that you succeed if you work hard.

Under the silent necessity modal account, we would expect it to be possible for negation to scope above the necessity operator, resulting in a weaker (and more plausible) reading: Hard work does not guarantee success. But the stronger reading is preferred.

Similar patterns obtain with downward entailing quantifiers and *only*.

(80) Nobody succeeds if they work hard.

(81) You succeed only if you work hard.

To deal with this data, some theorists depart from the silent necessity operator analysis. For example, Herburger (2015) argues that bare conditionals are ambiguous between an existential and a universal semantics, and Bassi & Bar-Lev (2018) argue for an unambiguously existential meaning that is strengthened to a universal-like

meaning by exhaustification (identically to the variable force modals from section 3.2).

Another approach is to adopt a non-quantificational semantics for bare conditionals (Schlenker 2004, Križ 2015). On the non-quantificational theory, bare conditionals are plural predication structures that exhibit homogeneity effects. The *if*-clause denotes a plurality of worlds, and the main clause proposition applies to the plurality. On this view, the distribution of (apparent) strong and weak readings can be explained as a consequence of homogeneity effects, parallel to definite plurals.

Concretely, (82-b) is treated as a negated plural predication structure, analogous to (83-b).

- (82) a. You succeed if you work hard.
- b. You don't succeed if you work hard.
- (83) a. Ann ate the cookies.
- b. Ann didn't eat the cookies.

The plural theory of bare conditionals has received renewed attention recently, and new empirical work aims to test its predictions (e.g. Marty et al. 2020).

Bare conditionals (especially in the present tense) have prominent readings that can be described as generic (von Stechow 1997). Generic sentences like those in (84) are descriptively characterized as sentences with a lawlike meaning. They can be used to convey claims about what is normal, typical, statistically regular, or required by relevant rules.

- (84) a. Ducks lay eggs.
- b. Bishops move diagonally.
- c. Adults eat arugula.
- d. Typhoons arise in this part of the Pacific.

Following the approach to quantificational adverbs taken by Lewis (1975), some work on the formal semantics of generics assumes the existence of a silent quantificational adverb called GEN, a universal quantifier restricted to certain worlds, situations, or cases (see e.g. Krifka et al. (1995) and references therein).

As with bare conditionals, the meanings of bare plural generic sentences under downward entailing operators are unexpectedly strong (given a universal semantics for GEN). For example, in (85), the reading seems to be that babies never eat arugula.

- (85) a. Babies don't eat arugula.
- b. Bill doesn't think babies eat arugula.

On the GEN analysis, wide-scope negation ought yield a reading closer to (86).

- (86) Babies don't always eat arugula.

But this kind of reading does not seem available for the sentences in (85).

To the extent that bare conditionals and bare plural generics share a common core of meaning, they may deserve a unified analysis. As with bare conditionals, there is a familiar space of options for the force of the GEN operator: It could be underlyingly existential and strengthened to a universal, it could be a necessity modal with additional domain restrictions, or it could have a non-quantificational semantics, perhaps based on plural reference to worlds, events, or situations.²⁰

For more on the semantics of generic sentences, see 6.6. GENERICITY AND MODALITY.

4.2 Imperatives

A significant line of work in semantics analyzes imperative clauses as modalized propositions.²¹ One straightforward way to implement this view is to assume that the logical form of an imperative clause contains a silent necessity modal.²² This approach straightforwardly captures the canonical cases, in which monoclausal imperatives are used to issue commands. However, there are cases where the modal force of imperatives seems to vary between necessity and possibility depending on the context.

In commands, the modal operator can be treated as having universal force. But, aside from issuing commands, imperatives may also be used to grant permission to perform the action described in the prejacent, or advise the addressee to do so.²³

- (87) a. Have a seat. (as an invitation)
b. Take a cookie. (offered, no obligation)

²⁰ See Križ (2015) for a suggestion along these lines.

21 See Kaufmann (2012) and references therein. Kaufmann presents various forms of evidence for this approach, including cases where *if*-clauses restrict imperatives, cases where imperatives appear to be nested under other modal operators, and apparent cases of modal subordination.

22 Two kinds of approaches to imperative semantics predominate: The propositional approach, where imperative clauses denote propositions, and the property approach, where imperative clauses denote properties. Here we focus on the propositional approach. For the property approach, see, for example, [Portner \(2007\)](#). In the property approach, questions about quantificational force do not usually arise, so we consider the property approach to be outside the scope of this article. (Though see [Roberts \(2023\)](#) for a hybrid analysis.)

23 Imperative clauses are not, strictly speaking, categorized as such by their discourse function, but they typically occur in monoclausal utterances used to issue commands, orders, or requests. Imperative morphosyntax varies from language to language, and may include null subjects, non-default word order, or tense-aspect-mood marking that is not found in declaratives.

Schwager (2005) presents examples of imperatives embedded under certain adverbials like *for example*, which have possibility-like readings.

- (88) A: How can I save money.
B: Stop buying cigarettes, for example.

If imperative clauses always contain a covert necessity modal, these possibility-like readings are unexpected. To deal with these cases, modal theories of imperatives must find ways to deliver readings that are weaker than strong necessity. Kaufmann (2012) argues that imperative clauses are ambiguous between a possibility reading and a necessity reading, where the latter is derived via exhaustification. Medeiros (2013) argues that imperative clauses convey a weak necessity meaning. Oikonomou (2016) argues for an underlying possibility meaning, which is strengthened via exhaustification.

Taking a step back, we can see this line of work as drawing an analogy between imperative clauses and variable force modals (see Sections 3.1-3.2). If this idea is correct, then variable force in imperatives may ultimately have the same source as variable force modality. For more on the meaning of imperatives, see Chapter 7.1. SPEECH ACTS AND MODALITY in this volume.

4.3 Infinitival clauses

Another apparent case of covert variable force modality comes from English infinitival relative clauses. In (89) below, the relative clause *for us to do* is ambiguous between at least two modal interpretations. Descriptively, its force can be either possibility-like or necessity-like.

- (89) Mrs. Schaden found many things for us to do.
a. Paraphrase 1: Mrs. Schaden found many things that we **could** do.
b. Paraphrase 2: Mrs. Schaden found many things that we **should** do.

Hackl & Nissenbaum (1999) attribute these modal interpretations to a covert modal operator inside the relative clause. They observe that the force of the covert modal depends on the determiner. When the determiner has a strong interpretation, the modal in the relative clause has a necessity reading. Otherwise, its strength may vary between necessity and possibility.

- (90) Determiner Modal Generalization Hackl & Nissenbaum (1999)
Strong determiners (and strong interpretations of weak determiners) always induce a *should*-reading in *for*-infinitival relative clauses. Weak interpretations of weak determiners allow both *could*- and *should*-readings.

The examples below illustrate the generalization, with weak determiners in (91) and strong determiners in (92).

- (91) Infinitival relatives under weak determiners. (✓ could, ✓ should)
 a. A man for John to play against is in the next room.
 b. Three men for John to play against are in the next room.
- (92) Infinitival relatives under strong determiners. (# could, ✓ should)
 a. The men for John to play against are in the next room.
 b. Every man for John to play against is in the next room.

Hackl & Nissenbaum (1999) established this generalization, but did not offer a full account. Bhatt (2006) presents a unified theory that aims to account for both (90) and other instances of covert modality in non-finite clauses, such as embedded questions like (93).²⁴

- (93) Tim knows how to solve the problem.

Infinitival relative clauses seem to present another case of covert variable force modality. For more on this topic, see 4.1. NONFINITE MODALITY.

5 Conclusion

In this chapter, we have looked at some cases that do not clearly fit into the opposition between universal and existential modal operators. As discussed in Section 2, weak necessity modals display a set of inference patterns that are not expected on a simple universal quantifier analysis. If weak necessity modals have the same flavor as their strong necessity counterparts, then the Kratzerian treatment of necessity modals must be amended in some way to capture the distinction between weak and strong necessity.

Variable force modals, whose meanings seem to depend on the the discourse context or the polarity of their environment, present another challenging case. The theories surveyed in Section 3 largely agree that each variable force modal has one underlying quantificational force, though they differ on other points, such as how the observed meanings are derived. Finally, we have also discussed the role of bare conditionals, generics, and imperatives, each of which has a profile that can be seen as analogous to weak necessity or variable force.

Our aim has been to point the reader to points of convergence in the field, as well as open questions. In particular, the sheer variety of theoretical approaches to

²⁴ See also Hackl & Nissenbaum (2012) for discussion of Bhatt's theory. More recently, Grano (2025) presents a theory of variable force in infinitival relatives based on degree semantics.

modal force raises the question of how modal meanings are learned. In addressing the acquisition question, theorists must consider which of these theories are in fact distinct hypotheses considered by learners. As we learn more about the typology of modal meanings, the field continues to make progress on these issues.

References

- Agha, Omar & Paloma Jeretič. 2022. Weak necessity modals as homogeneous pluralities of worlds. In *Proceedings of SALT 32*, vol. 32, .
- Van der Auwera, Johan & Vladimir A Plungian. 1998. Modality's semantic map. *Linguistic typology* 2(1). 79–124.
- Bar-Lev, Moshe E & Danny Fox. 2020. Free choice, simplification, and innocent inclusion. *Natural Language Semantics* 28(3). 175–223.
- Bassi, Itai & Moshe E. Bar-Lev. 2018. A unified existential semantics for bare conditionals. *Proceedings of Sinn und Bedeutung* 21(1). 125–142.
- Bhatt, Rajesh. 2006. *Covert modality in non-finite contexts*. De Gruyter Brill.
- Bian, Mingyang & Paloma Jeretič. in prep. The semantic development of the Chinese modal light verb DE.
- Bochnak, M Ryan. 2015a. Underspecified modality in Washo. In *Proceedings of the Workshop on Structure and Constituency in Languages of the Americas*, vol. 18, 3–17.
- Bochnak, M Ryan. 2015b. Variable force modality in Washo. In *Proceedings of North-East Linguistics Society*, vol. 45, 105–114.
- Chierchia, Gennaro, Danny Fox & Benjamin Spector. 2012. The grammatical view of scalar implicatures and the relationship between semantics and pragmatics. *Semantics: An international handbook of natural language meaning* 3. 2297–2332.
- Davis, Henry, Lisa Matthewson & Hotze Rullmann. 2009. Out of control marking as circumstantial modality in St'át'imcets. *Cross-linguistic semantics of tense, aspect and modality* 205–244.
- Deal, Amy Rose. 2011. Modals without scales. *Language* 559–585.
- von Fintel, Kai & Sabine Iatridou. 2023. Prolegomena to a theory of X-Marking. *Linguistics and Philosophy* 46(6). 1467–1510. <https://doi.org/10.1007/s10988-023-09390-5>.
- Fox, Danny. 2007. Free choice and the theory of scalar implicatures. In *Presupposition and implicature in compositional semantics*, 71–120. Springer.
- Gajewski, Jon Robert. 2005. *Neg-raising: Polarity and presupposition*. Massachusetts Institute of Technology Thesis.
- Grano, Thomas. 2025. Variable-force modality in english infinitival relatives: A matter of degree. *Linguistic Inquiry* 1–33. https://doi.org/10.1162/ling_a_00542.

- https://doi.org/10.1162/ling_a_00542.
- Grubic, Mira & Anne Mucha. 2021. Decomposing necessity—The Hausa exclusive particle *sai* as a window into the building blocks of modal meaning. In *Proceedings of sinn und bedeutung*, vol. 25, 358–375.
- Guo, Qingxia, Nathaniel Imel & Shane Steinert-Threlkeld. 2022. A database for modal semantic typology. In *Proceedings of the 4th workshop on research in computational linguistic typology and multilingual nlp*, 42–51.
- Hackl, Martin & Jon Nissenbaum. 1999. Variable modal force in for-infinitival relative clauses. In *The proceedings of the bergamo conference on the syntax and semantics of tense and mood selection*, .
- Hackl, Martin & Jon Nissenbaum. 2012. A modal ambiguity in for-infinitival relative clauses. *Natural Language Semantics* 20(1). 59–81.
- Herburger, Elena. 2015. Only if: If only we understood it. *Proceedings of Sinn und Bedeutung* 19. 304–321. <https://doi.org/10.18148/sub/2015.v19i0.235>.
- Herburger, Elena. 2019. Bare conditionals in the red. *Linguistics and Philosophy* 42(2). 131–175. <https://doi.org/10.1007/s10988-018-9242-2>.
- Hohaus, Vera, Siena Weingartz & Marijn Boomars. 2023. Variable strength in necessity modality: A case of variation between Afrikaans, Dutch, English, and German. *Festschrift for Hotze Rullmann (to appear)*. *University of British Columbia Working Papers in Linguistics (UBCWPL)* .
- Izvorski, Roumyana. 1997. The Present Perfect as an Epistemic Modal. In *Semantics and Linguistic Theory*, vol. 7, 18.
- Jeretič, Paloma. 2021a. *Neg-raising modals and scaleless implicatures*: New York University dissertation.
- Jeretič, Paloma. 2021b. Scaleless modals in Ecuadorian Siona and grammaticalized scalar reasoning. *Proceedings of Semantics of Understudied Languages in the Americas* 11.
- Kaufmann, Magdalena. 2012. *Interpreting Imperatives*. Springer Dordrecht.
- Kiefer, Ferenc. 1986. Focus and modality. In *Linguistic and Literary Studies in Eastern Europe*, John Benjamins.
- Kratzer, Angelika. 1977. What ‘must’ and ‘can’ must and can mean. *Linguistics and Philosophy* 1(3). 337–355. <https://doi.org/10.1007/BF00353453>.
- Kratzer, Angelika. 1981. The notional category of modality .
- Kratzer, Angelika. 1986. Conditionals. *Chicago Linguistics Society* 22(2). 1–15.
- Kratzer, Angelika. 1991. *Modality. Semantics: An international handbook of contemporary research*. Berlin: De Gruyter.
- Kratzer, Angelika. 2021. Chasing hook: Quantified indicative conditionals. In Lee Walters & John Hawthorne (eds.), *Conditionals, paradox, and probability: Themes from the philosophy of dorothy edgington*., Oxford University press.

- Krifka, Manfred, Francis Jeffry Pelletier, Gregory Carlson, Alice ter Meulen, Godehard Link & Gennaro Chierchia. 1995. Genericity: An introduction. In Greg Carlson & Francis Jeffry Pelletier (eds.), *The Generic Book*, .
- Križ, Manuel. 2015. *Aspects of Homogeneity in the Semantics of Natural Language*: University of Vienna dissertation.
- Križ, Manuel. 2016. Homogeneity, Non-Maximality, and all. *Journal of Semantics* 33(3). 493–539. <https://doi.org/10.1093/jos/ffv006>.
- Lassiter, Daniel. 2011. *Measurement and modality: The scalar basis of modal semantics*: New York University dissertation.
- Leffel, Timothy. 2012. Modality and future reference in masalit. In *Selected Proceedings of the 42nd Annual Conference on African Linguistics: African Languages in Context*, .
- Lewis, David K. 1975. Adverbs of quantification. In Edward Louis Keenan (ed.), *Formal semantics of natural language: papers from a colloquium sponsored by the King's College Research Centre, Cambridge*, 3–15. Cambridge University Press.
- Löbner, Sebastian. 1985. Definites. *Journal of Semantics* 4(4). 279–326. <https://doi.org/10.1093/jos/4.4.279>.
- Marty, Paul, Jacopo Romoli & Paolo Santorio. 2020. Counterfactuals and undefinedness: Homogeneity vs supervaluations. In *Proceedings of SALT 30*, .
- Matthewson, Lisa. 2013. Gitksan modals. *International Journal of American Linguistics* 79(3). 349–394.
- Medeiros, David J. 2013. *Formal approaches to the syntax and semantics of imperatives*: University of Michigan dissertation.
- Menzies, Stacey. 2013. *Nsyilxcen modality: Semantic analysis of epistemic modality*: University of British Columbia dissertation.
- Nauze, Fabrice. 2008. *Modality in typological perspective*: University of Amsterdam dissertation.
- Newkirk, Lydia. 2022a. *Be flexible, but not too flexible: Limited variable-force modals in kinande and the typology of modal force*: Rutgers dissertation.
- Newkirk, Lydia. 2022b. Limited variable-force modals and the interactions of scal {ar, eless} implicatures. In *Proceedings of Sinn und Bedeutung*, vol. 26, 641–657.
- Oikonomou, Despina. 2016. Imperatives are existential modals: Deriving the strong reading as an implicature. In *Proceedings of SALT 26*, .
- Peterson, Tyler Roy Gösta. 2010. *Epistemic modality and evidentiality in gitksan at the semantics-pragmatics interface*: University of British Columbia dissertation.
- Portner, Paul. 2007. Imperatives and modals. *Natural Language Semantics* 15(4). 351–383. <https://doi.org/10.1007/s11050-007-9022-y>.

- Roberts, Craige. 2023. Imperatives in a dynamic pragmatics. *Semantics and Pragmatics* 16.
- Rubinstein, Aynat. 2014. On Necessity and Comparison. *Pacific Philosophical Quarterly* 95(4). 512–554. <https://doi.org/10.1111/papq.12047>.
- Rubinstein, Aynat. 2020. Weak necessity. *The Wiley Blackwell companion to semantics*.
- Rullmann, Hotze, Lisa Matthewson & Henry Davis. 2008. Modals as distributive indefinites. *Natural Language Semantics* 16(4). 317–357. <https://doi.org/10.1007/s11050-008-9036-0>.
- Schlenker, Philippe. 2004. Conditionals as Definite Descriptions. *Research on Language and Computation* 2(3). 417–462. <https://doi.org/10.1007/s11168-004-0908-2>.
- Schmitt, Viola. 2023. Are there pluralities of worlds? *Journal of Semantics* 40(1). 153–178. <https://doi.org/10.1093/jos/ffad002>. <https://doi.org/10.1093/jos/ffad002>.
- Schwager, Magdalena. 2005. Exhaustive imperatives. In *Proceedings of amsterddam colloquium*, .
- Silk, Alex. 2022. Weak and strong necessity modals: On linguistic means of expressing "a primitive concept ought". In Billy Dunaway & David Plunkett (eds.), *Meaning, Decision, and Norms: Themes From the Work of Allan Gibbard*, 203–245. Maize Books.
- Staniszewski, Frank. 2019. Wanting, acquiescing, and neg-raising. In Duygu Göksu & Jonathan Pesetsky (ed.), *Proceedings of the North East Linguistics Society* 49, Amherst, MA: GLSA.
- Uegaki, Wataru, Anne Mucha, Ella Hannon, James Engels & Fred Whibley. 2025. Cross-linguistic dataset of force-flavor combinations in modal elements. *Linguistic Variation* 25(2). 269–305.
- Vander Klok, Jozina & Vera Hohaus. 2020. Weak necessity without weak possibility: The composition of modal strength distinctions in Javanese. *Semantics and Pragmatics* 13(0). 12. <https://doi.org/10.3765/sp.13.12>.
- von Fintel, Kai. 1997. Bare Plurals, Bare Conditionals, and Only. *Journal of Semantics* 14(1). 1–56. <https://doi.org/10.1093/jos/14.1.1>.
- von Fintel, Kai & Sabine Iatridou. 2008. How to Say Ought in Foreign: The Composition of Weak Necessity Modals. In Jacqueline Guéron & Jacqueline Lecarme (eds.), *Time and Modality Studies in Natural Language and Linguistic Theory*, 115–141. Dordrecht: Springer Netherlands. https://doi.org/10.1007/978-1-4020-8354-9_6.
- Weingartz, Siena & Vera Hohaus. 2024. Variable modal strength in Afrikaans and Samoan: Deriving strong necessity from weak necessity. In *Proceedings of TripleA 10*, Universität Potsdam.

Yanovich, Igor. 2016. Old english *motan, variable-force modality, and the presupposition of inevitable actualization. *Language* 92(3). 489–521.